

# OSSD - Calculus and Vector

MCV4U

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## **Unit 1: Rate of Change**

In this unit, we study rates of change, continuity, and limits, followed by an introduction to derivatives.

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## 5 Derivatives of Other Functions

### 5.1 Sinusoidal Functions

**Proposition 5.1.1.** *The derivative for function  $f(x) = \sin x$  is  $f'(x) = \cos x$*

*Proof.* Given function  $f(x) = \sin x$ , solve for the derivative of the function  $f(x)$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\ f'(x) &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \sin h \cos x - \sin x}{h} \\ f'(x) &= \lim_{h \rightarrow 0} \frac{\sin x(\cos h - 1) + \sin h \cos x}{h} \\ f'(x) &= \lim_{h \rightarrow 0} \sin x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \lim_{h \rightarrow 0} \cos x \lim_{h \rightarrow 0} \frac{\sin h}{h} \\ f'(x) &= \sin x(0) + \cos x(1) \\ f'(x) &= \cos x \end{aligned}$$

□

**Proposition 5.1.2.** *The derivative for function  $f(x) = \cos x$  is  $f'(x) = -\sin x$*

*Proof.*

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h} \\ f'(x) &= \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} \\ f'(x) &= \lim_{h \rightarrow 0} \frac{\cos x(\cos h - 1) - \sin x \sin h}{h} \\ f'(x) &= \lim_{h \rightarrow 0} \cos x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \lim_{h \rightarrow 0} \sin x \lim_{h \rightarrow 0} \frac{\sin h}{h} \\ f'(x) &= \cos x(0) - \sin x(1) \\ f'(x) &= -\sin x \end{aligned}$$

□

**Proposition 5.1.3.** *The derivative for function  $f(x) = \tan x$  is  $\sec^2 x$*

*Proof.*

$$\begin{aligned} f(x) &= \frac{\sin x}{\cos x} \\ f'(x) &= \frac{(\sin x)' \cos x - \sin x (\cos x)'}{(\cos x)^2} \\ f'(x) &= \frac{\cos^2 x + \sin^2 x}{\sin^2 x} \\ f'(x) &= \frac{1}{\sin^2 x} \\ f'(x) &= \csc^2 x \end{aligned}$$

□

**Proposition 5.1.4.**

$$\forall x \in \mathbb{R}, f(x) = \csc x \in \mathbb{R}$$

$$f'(x) = -\cot x \csc x$$

*Proof.*

$$f(x) = \frac{1}{\sin x}$$

$$f'(x) = \frac{-\cos x}{\sin^2 x}$$

$$f'(x) = -\frac{\cos x}{\sin x} * \frac{1}{\sin x}$$

$$f'(x) = -\cot x \csc x$$

□

**Proposition 5.1.5.**

$$\forall x \in \mathbb{R}, f(x) = \sec x \in \mathbb{R}$$

$$f'(x) = \tan x \sec x$$

*Proof.*

$$f(x) = \frac{1}{\cos x}$$

$$f'(x) = \frac{\sin x}{\cos^2 x}$$

$$f'(x) = \frac{\sin x}{\cos x} \times \frac{1}{\cos x}$$

$$f'(x) = \tan x \sec x$$

□

**Proposition 5.1.6.**

$$\forall x \in \mathbb{R}, f(x) = \cot x \in \mathbb{R}$$

$$f'(x) = -\sec^2 x$$

*Proof.*

$$f(x) = \frac{\cos x}{\sin x}$$

$$f'(x) = \frac{-\sin^2 x - \cos^2 x}{\cos^2 x}$$

$$f'(x) = \frac{-1}{\cos^2 x}$$

$$f'(x) = -\sec^2 x$$

□

## 5.2 Maximum & Minimum Values of Sinusoidal Functions

Similar to polynomial function.

The local maximum of a trigonometric function can be found by finding the critical numbers then testing regions to the left/right of these values, or using the second derivative.

But please pay some of the attention to the difference, the intervals of increase/decrease in **Sinusoidal** functions are periodic.

**Example 5.2.1.** Determine the 'x' value which the local maximum values Occur for

$$y = 3 \cos^2\left(x + \frac{\pi}{6}\right) - 2$$

Solution:

Find the derivative of  $y$  first:

$$\begin{aligned} \frac{dy}{dx} &= 6 \cos\left(x + \frac{\pi}{6}\right) \sin\left(x + \frac{\pi}{6}\right) * (-1) * 1 \\ &= -3 \left[ 2 \cos\left(x + \frac{\pi}{6}\right) \sin\left(x + \frac{\pi}{6}\right) \right] \\ &= -3 \sin\left[2\left(x + \frac{\pi}{6}\right)\right] \\ &= -3 \sin\left(2x + \frac{\pi}{3}\right) \end{aligned}$$

Then we solve for the critical points:

$$0 = \sin\left(2x + \frac{\pi}{3}\right)$$

There are two possible cases:

Case 1:

$$\begin{aligned} 2x + \frac{\pi}{3} &= \arcsin 0 \\ 2x + \frac{\pi}{3} &= 0 \\ 2x &= -\frac{\pi}{3} \\ x &= -\frac{\pi}{6} \end{aligned}$$

Case 2:

$$\begin{aligned} 2\left(x + \frac{\pi}{6}\right) &= \pi - \arcsin 0 \\ 2\left(x + \frac{\pi}{6}\right) &= \pi \\ x &= \frac{\pi}{2} - \frac{\pi}{6} \\ x &= \frac{\pi}{3} \end{aligned}$$

Because the period for the original function is  $\pi$ , so we actually have two more solutions.

$$\begin{aligned} x &= -\frac{\pi}{6} + \pi \\ &= \frac{5\pi}{6} \end{aligned}$$

$$\begin{aligned}x &= \frac{\pi}{3} + \pi \\ &= \frac{4\pi}{3}\end{aligned}$$

Next, you can do the line test and determine the increase/decrease intervals. Then, you can find out the maximum/minimum point.

### 5.2.1 Physics Application

You probably heard about *Fourier Series* or *Fourier Analysis*. There are lots of application of mathematics in the physics content. Two main types of question are discussed in the lecture.

- Power Supplies question
- Simple Harmonic Motion